On quadratic residue codes and hyperelliptic curves

Revised and updated

David Joyner* Claude†

November 2026

Abstract

This AI-aided paper revises and updates the first author's earlier paper [18]. For an odd prime p and a non-empty subset $S \subset \mathbb{F}_p$, let X_S be the hyperelliptic curve $y^2 = f_S(x)$, where $f_S(x) = \prod_{a\in S}(x-a)$. The quasi-quadratic residue (QQR) code C_{NQ} is a binary double-circulant code of length $2p$ whose codeword weights are given exactly by the numbers $\#X_T(\mathbb{F}_p)$ for sets T of even size. We use this dictionary in both directions. Coding-theory bounds give point-count bounds: for every $c < 1.6258$ and every sufficiently large prime p there is a set S with $\#X_S(\mathbb{F}_p) > cp$; if $p \equiv 3 \pmod 4$ and $p \geq 11$ there is a set with $\#X_S(\mathbb{F}_p) \geq \frac{5}{3}p - 6$. Conversely, if for some $c < 1.7799$ every such curve had at most cp points for infinitely many p, Goppa's conjecture would fail. We prove that C_{NQ} is self-dual for $p \equiv 3 \pmod 4$ and has complementary dual for $p \equiv 1 \pmod 4$, and that its extension by the all-ones vector is isodual. A quadratic-twist identity, $\#X_T + \#X_{T'} = 2p+2$, pairs curves with many and with few points. We record explicit constructions (due to Voloch) of sets with about $1.5p$ points for every p, and give examples of optimal-looking codes whose Duursma zeta functions violate the analog of the Riemann hypothesis.

Keywords: binary linear codes, double circulant codes, quadratic residue codes, hyperelliptic curves over finite fields, Legendre sums, Duursma zeta function.
MSC 2020: 94B05, 94B65, 11T71, 11T24, 14G50.

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*wdjoyner@gmail.com

†An AI model developed by Anthropic (Claude Opus 5.5). Contributions to this revision are described in the Acknowledgements and in the accompanying revision log.

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1 Introduction

This section sets the scene. We recall the coding-theory and curve-counting background needed to state our results, formulate the two conjectures that motivate the paper (Goppa's and Bazzi–Mitter's), summarize what we prove, and pose the two questions that the later sections address.

Finding explicit families of binary linear codes that are asymptotically good, let alone ones that beat the Gilbert–Varshamov bound, is a long-standing problem. This paper studies one attack on it, through a dictionary between certain double-circulant codes built from quadratic residues and the number of points on hyperelliptic curves over a prime field. Codes with this kind of relationship were studied by Helleseth [14], Bazzi and Mitter [2], Voloch [32], and Helleseth and Voloch [15].

1.1 Codes

Here we fix the basic vocabulary of linear codes and state the Gilbert–Varshamov bound, which is the benchmark every asymptotic result below is measured against.

A binary linear $[n,k,d]$ code is a k-dimensional subspace $C \subseteq \mathbb{F}_2^n$ whose smallest nonzero Hamming weight is d. Writing $V(n,r)$ for the size of a Hamming ball of radius r, the Gilbert–Varshamov (GV) bound guarantees an $[n,k,d]$ code with $k \geq n - \log_2 V(n,d-1)$. Asymptotically:

for each $0 < \delta < 1/2$ there are codes with relative distance $d/n \to \delta$ and rate $k/n \to 1 - H(\delta)$, where

$$H(\delta) = -\delta \log_2 \delta - (1-\delta)\log_2(1-\delta)$$

is the binary entropy function.

Conjecture 1.1 (Goppa [12])**.** *The asymptotic binary GV bound is tight: if $\alpha(\delta)$ denotes the largest asymptotic rate of binary codes with relative distance at least δ, then $\alpha(\delta) = 1 - H(\delta)$ for $0 \le \delta \le 1/2$.*

1.2 Curves

Here we introduce the curves X_S attached to subsets S of $\mathbb{F}_p$ and the statement $B(c,p)$, an upper bound on their point counts that plays the role of a minimum-distance bound throughout the paper.

For a polynomial $h \in \mathbb{F}_p[x]$ with distinct roots, $X : y^2 = h(x)$ denotes the smooth projective curve with that affine equation, and $\#X(\mathbb{F}_p)$ its number of $\mathbb{F}_p$-rational points (affine points plus the points at infinity). For each non-empty $S \subseteq \mathbb{F}_p$ put

$$f_S(x) = \prod_{a \in S} (x - a), \qquad X_S : y^2 = f_S(x), \qquad \#X_S = \#X_S(\mathbb{F}_p).$$

Weil's theorem gives $\#X_S \approx p$ when $|S|$ is small compared to $\sqrt{p}$, but says nothing when $|S|$ is a positive proportion of p. Trivially $\#X_S \le 2p+1$.

Definition 1.2. For $c > 0$, let $B(c,p)$ be the statement: $\#X_S \le cp$ *for every non-empty* $S \subseteq \mathbb{F}_p$. Let $B_{\mathrm{ev}}(c,p)$ be the same statement restricted to sets S of even size.

Conjecture 1.3 (Bazzi–Mitter [2])**.** *There is $c \in (0,2)$ such that $B(c,p)$ holds for infinitely many primes p.*

1.3 Results

This subsection lists the main results, with pointers to where each is proved. The reader who wants only the statements can stop here and turn to Section 7.

The QQR code $C_{NQ} \subseteq \mathbb{F}_2^{2p}$ (Section 3) has dimension p or $p-1$, and Proposition 3.6 expresses each codeword weight as $2p+2-\#X_T$ or $\#X_T - 2$ for a set T of even size. Our main results are:

1. C_{NQ} is self-dual if $p \equiv 3 \pmod 4$ and LCD (linear complementary dual) if $p \equiv 1 \pmod 4$; its extension by the all-ones word is isodual when $p \equiv 1 \pmod 4$ (Theorem 3.4, Theorem 6.2).

2. A quadratic-twist identity: for T of even size and suitable $a \notin T$, $\#X_T + \#X_{T'} = 2p+2$ for an explicit set T' of the same size (Lemma 2.4).

3. If $B_{\mathrm{ev}}(c,p)$ holds for infinitely many primes p with $c < 2(1 - H^{-1}(1/2)) \approx 1.7799$, then Goppa's conjecture is false (Theorem 4.1). No congruence condition on p is needed.

4. For every $c < 1.6258$ and all sufficiently large p, there is S with $\#X_S > cp$ (Theorem 4.3). The proof is non-constructive.

5. If $p \equiv 3 \pmod 4$ and $p \geq 11$, there is S with $\#X_S \geq \frac{5}{3}p - 6$ (Corollary 4.5).

6. Explicitly, for every prime $p \geq 11$ the set $S = Q$, N or $Q \cup \{0\}$ (depending on p mod 8) has $\#X_S \geq 1.5p - \frac{1}{2}$ (Theorem 5.1, due to Voloch, with the case $p \equiv 5 \pmod 8$ added).

7. The Duursma zeta function of C_{NQ} satisfies the Riemann hypothesis for $p = 7, 11, 19$ but not for $p = 23$, and that of the extended code fails for $p = 13, 17$ (Section 6).

1.4 Questions

Two natural questions arise from the results above: one asks for explicit curves with many points, and the other asks whether point counts are bounded uniformly below $2p$.

Question 1.4. For each large prime p, is there an *effectively computable* $S \subseteq \mathbb{F}_p$ with $\#X_S > cp$ for a fixed $c > 1$?

Question 1.5. Is there $c < 2$ such that $\#X_S < cp$ for all sufficiently large p and all $S \subseteq \mathbb{F}_p$?

Item 6 above answers Question 1.4 affirmatively for $c = 1.5 - o(1)$. Voloch's Chebotarev construction (Theorem 5.3) answers Question 1.5 negatively. Section 7 discusses what remains open and how hard it appears.

2 Notation and background

This section collects the tools used in the rest of the paper. We first describe arithmetic in the group algebra $\mathbb{F}_2[x]/(x^p - 1)$, where the codes live. Next we relate the point count of X_S to a Legendre character sum. Finally we prove a quadratic-twist identity that pairs curves with many points and curves with few.

Throughout, p is an odd prime, $Q \subset \mathbb{F}_p^\times$ the set of nonzero squares, $N = \mathbb{F}_p^\times \setminus Q$ the non-squares, and $\chi = \left(\frac{\cdot}{p}\right)$ the Legendre symbol, with $\chi(0) = 0$. Recall $-1 \in Q$ iff $p \equiv 1 \pmod 4$ and $2 \in Q$ iff $p \equiv \pm 1 \pmod 8$.

2.1 Cyclotomic arithmetic modulo 2

Subsets of $\mathbb{F}_p$ are encoded as elements of a binary group algebra. This subsection lists the few identities in that algebra that we need, chiefly a formula for the weight of a product, (2).

Let $R = \mathbb{F}_2[x]/(x^p - 1)$. For $S \subseteq \mathbb{F}_p$ put $r_S = \sum_{i \in S} x^i \in R$ (so $r_\emptyset = 0$), and let $J = r_{\mathbb{F}_p} = 1 + x + \cdots + x^{p-1}$. We identify $u \in R$ with its coefficient vector in $\mathbb{F}_2^p$, so $\mathrm{wt}(r_S) = |S|$. Then:

- $1 + r_Q + r_N = J$, and $J^2 = pJ = J$;
- $r_S J = |S| J$, so $\mathrm{wt}(u) \bmod 2$ is the ring homomorphism $u \mapsto u(1)$;
- $\mathrm{wt}(r_A r_B) \equiv |A|\,|B| \pmod 2$;
- $r_S^2 = r_{2S}$; in particular $r_Q^2 = r_Q$ iff $2 \in Q$.

For $u \in R$ let $u^-(x) = u(x^{-1})$. Then $u \mapsto u^-$ is a ring automorphism, $r_S^- = r_{-S}$, and so

$$r_Q^- = r_Q,\ r_N^- = r_N \ \ (p \equiv 1 \bmod 4), \qquad r_Q^- = r_N,\ r_N^- = r_Q \ \ (p \equiv 3 \bmod 4). \tag{1}$$

The standard inner product on $\mathbb{F}_2^p$ is $\langle u, v\rangle = \mathrm{ct}(u v^-)$, where ct denotes the constant term.

The coefficient of x^a in $r_A r_B$ is the parity of $|A \cap (a - B)|$, where $a - B = \{a - b : b \in B\}$. Hence

$$\mathrm{wt}(r_A r_B) = \sum_{a \in \mathbb{F}_p} \pi\big(|A \cap (a - B)|\big), \qquad \pi(m) = m \bmod 2 \in \{0,1\}. \tag{2}$$

Example 2.1. Let $p = 7$, so $Q = \{1,2,4\}$ and $N = \{3,5,6\}$. With $S = \{0,1\}$, $r_S = 1 + x$ and

$$r_N r_S = 1 + x^3 + x^4 + x^5, \qquad r_Q r_S = x + x^3 + x^4 + x^5.$$

Each has weight 4, in agreement with $\mathrm{wt}(r_A r_B) \equiv |A||B| = 6 \equiv 0 \pmod 2$.

2.2 Point counts and character sums

The number of points on X_S is p plus a correction given by a sum of Legendre symbols. This subsection states that relation, Weil's bound on the sum, and the trivial bounds that hold when $|S|$ is large.

Lemma 2.2 (Artin, Hasse, Weil)**.** *Let $S \subseteq \mathbb{F}_p$ be non-empty and $\Sigma_S = \sum_{x \in \mathbb{F}_p} \chi(f_S(x))$.*

1. *If $|S|$ is even, $\#X_S = p + 2 + \Sigma_S$, and X_S has genus $g = (|S| - 2)/2$.*
2. *If $|S|$ is odd, $\#X_S = p + 1 + \Sigma_S$, and X_S has genus $g = (|S| - 1)/2$.*
3. *In both cases $|\#X_S - (p+1)| \le 2g\sqrt{p}$.*

Proof. Each $x \in \mathbb{F}_p$ contributes $1 + \chi(f_S(x))$ affine points. Because f_S is monic, the smooth model has two rational points at infinity when $\deg f_S$ is even and one when it is odd. Part 3 is Weil's theorem [34]; see [28, §2.11]. □

Part 3 is only informative when $|S|$ is small compared with $\sqrt{p}$. Note also the trivial bounds, valid for all non-empty S:

$$|S| + 1 \ \le \ \#X_S \ \le \ 2p - |S| + 2. \tag{3}$$

Example 2.3. Take $p = 7$ and $S = \{0,1\}$, so $f_S(x) = x(x-1)$. The values $\chi(f_S(x))$ for $x = 0, 1, \ldots, 6$ are $0, 0, 1, -1, -1, -1, 1$. So $\Sigma_S = -1$ and $\#X_S = 7 + 2 - 1 = 8$. The curve has genus 0 and indeed $\#X_S = p + 1$.

2.3 A quadratic-twist identity

A change of variables $x = a + 1/u$ turns X_T into another curve of the same shape, or into its quadratic twist. The resulting identity pairs curves with many points and curves with few points. It is the key new ingredient in Corollary 4.5.

Lemma 2.4 (Twist identity)**.** *Let $T \subseteq \mathbb{F}_p$ be non-empty of even size, let $a \in \mathbb{F}_p \setminus T$, and put*

$$\varepsilon = \chi(f_T(a)) \in \{\pm 1\}, \qquad T' = \{(t-a)^{-1} : t \in T\}.$$

Then $|T'| = |T|$, and

$$\#X_T = \#X_{T'} \ \text{ if } \varepsilon = 1, \qquad \#X_T + \#X_{T'} = 2p+2 \ \text{ if } \varepsilon = -1.$$

Proof. For $u \neq 0$ write $x = a + u^{-1}$. Then $x - t = u^{-1}\big((a-t)u + 1\big) = u^{-1}(a-t)\big(u - (t-a)^{-1}\big)$, so

$$f_T(a+u^{-1}) = u^{-|T|}\, f_T(a)\, f_{T'}(u).$$

As $|T|$ is even, $\chi(f_T(a+u^{-1})) = \varepsilon\,\chi(f_{T'}(u))$. The map $u \mapsto a + u^{-1}$ is a bijection $\mathbb{F}_p^{\times} \to \mathbb{F}_p \setminus \{a\}$. Also $f_{T'}(0) = \prod_t (t-a)^{-1} = f_T(a)^{-1}$ (as $|T|$ is even), so $\chi(f_{T'}(0)) = \varepsilon$. Hence

$$\Sigma_T = \varepsilon + \varepsilon\big(\Sigma_{T'} - \varepsilon\big) = \varepsilon\,\Sigma_{T'} + \varepsilon - 1.$$

By Lemma 2.2(1), $\#X_T = p + 2 + \Sigma_T$ and $\#X_{T'} = p + 2 + \Sigma_{T'}$. If $\varepsilon = 1$ this gives $\#X_T = \#X_{T'}$; if $\varepsilon = -1$ it gives $\#X_T = p - \Sigma_{T'} = 2p + 2 - \#X_{T'}$. □

Geometrically, X_T is isomorphic to $w^2 = \varepsilon f_{T'}(u)$, which is $X_{T'}$ or its quadratic twist.

Example 2.5. Let $p = 11$ and $T = \{1,2,3,4\}$. Here $f_T(0) = 24 \equiv 2$ and $\chi(2) = -1$ since $11 \equiv 3 \pmod 8$. With $a = 0$ we get $T' = \{1,6,4,3\} = \{1,3,4,6\}$. Direct counting gives $\#X_T = 8$ and $\#X_{T'} = 16$, and $8 + 16 = 24 = 2p + 2$.

For p odd put

$$M_p = \max\{\#X_T : \ T \subseteq \mathbb{F}_p,\ |T| \text{ even},\ T \neq \emptyset\}, \qquad m_p = \min\{\#X_T : \ T \subseteq \mathbb{F}_p,\ |T| \text{ even},\ T \neq \emptyset\}.$$

Corollary 2.6. *For every odd prime p, $M_p + m_p \geq 2p + 2$.*

Proof. Let T attain m_p. If $\chi(f_T(a)) = -1$ for some $a \notin T$, then Lemma 2.4 gives $M_p \geq \#X_{T'} = 2p + 2 - m_p$. Otherwise $\chi(f_T(a)) = 1$ for all $a \notin T$, so $\Sigma_T = p - |T|$ and $m_p = \#X_T = 2p + 2 - |T| \geq p + 3$. Then $M_p \geq m_p \geq p + 3$ and $M_p + m_p \geq 2p + 6$. □

3 Quasi-quadratic residue codes

This section defines the quasi-quadratic residue code C_{NQ} and develops its structure. We compute its dimension and determine its dual. We then prove the central dictionary (Proposition 3.6), which turns each codeword weight into a point count. Finally we single out the even subcode C_{ev}, on which all the later bounds rest, and tabulate weight distributions.

Definition 3.1. The *quasi-quadratic residue code* (QQR code) is

$$C_{NQ} = \{\, c_S = (r_N r_S,\ r_Q r_S) : \ S \subseteq \mathbb{F}_p \,\} \subseteq R \oplus R \cong \mathbb{F}_2^{2p}.$$

Equivalently, C_{NQ} is the image of the $\mathbb{F}_2$-linear map $\varphi : R \to R \oplus R$, $u \mapsto (r_N u, r_Q u)$. It is a pure double-circulant code (a quasi-cyclic code of index 2) with generator matrix $[\,\mathrm{circ}(r_N) \mid \mathrm{circ}(r_Q)\,]$.

Proposition 3.2 (Dimension)**.** $\ker\varphi = \{0\}$ *if* $p \equiv 3 \pmod 4$ *and* $\ker\varphi = \{0, J\}$ *if* $p \equiv 1 \pmod 4$. *Hence* $\dim C_{NQ} = p$ *or* $p-1$ *respectively. Every codeword has even weight.*

Proof. If $r_N u = r_Q u = 0$ then $0 = (r_N + r_Q)u = (1+J)u = u + \mathrm{wt}(u)J$, so $u \in \{0, J\}$. Moreover $\varphi(J) = (|N|J, |Q|J)$ vanishes iff $|N| = (p-1)/2$ is even, that is, iff $p \equiv 1 \pmod 4$. Finally $\mathrm{wt}(c_S) \equiv |N||S| + |Q||S| = (p-1)|S| \equiv 0 \pmod 2$. □

Example 3.3. For $p = 5$ the codeword $c_{\mathbb{F}_5} = (r_N J, r_Q J) = (2J, 2J) = 0$, so $\dim C_{NQ} = 4$. For $p = 7$, $c_{\mathbb{F}_7} = (3J, 3J) = (J, J)$ is the all-ones word of length 14, and $\dim C_{NQ} = 7$.

3.1 Duality

The behavior of C_{NQ} under duality depends on whether -1 is a square modulo p. When it is not, the code is self-dual, which brings in the strong bounds for self-dual codes. When it is, the code meets its dual only in 0.

Theorem 3.4. *1. If* $p \equiv 3 \pmod 4$*, then* C_{NQ} *is a self-dual* $[2p, p]$ *code. It contains the all-ones word, and* $\mathrm{wt}(c_S) \equiv 2|S| \pmod 4$ *for every* S*. In particular* C_{NQ} *is singly-even (Type I), and its doubly-even subcode is* $\{c_S : |S| \text{ even}\}$.

2. If $p \equiv 1 \pmod 4$*, then* $C_{NQ} \cap C_{NQ}^{\perp} = \{0\}$*, so* $\mathbb{F}_2^{2p} = C_{NQ} \oplus C_{NQ}^{\perp}$*; that is,* C_{NQ} *is an LCD code.*

Proof. For $u, v \in R$,

$$\langle \varphi(u), \varphi(v) \rangle = \mathrm{ct}\left(u v^{-} (r_N r_N^{-} + r_Q r_Q^{-})\right). \tag{4}$$

(1) By (1), $r_N r_N^{-} + r_Q r_Q^{-} = 2 r_N r_Q = 0$, so C_{NQ} is self-orthogonal; it has dimension p, so it is self-dual. The all-ones word is $c_{\mathbb{F}_p}$ (Example 3.3). In a self-orthogonal binary code $\mathrm{wt}(x+y) = \mathrm{wt}\,x + \mathrm{wt}\,y - 2\,\mathrm{wt}(x \wedge y)$ with $\mathrm{wt}(x \wedge y)$ even, so weight modulo 4 is additive. Since $c_S = \sum_{i \in S} c_{\{i\}}$ and $\mathrm{wt}(c_{\{i\}}) = |N| + |Q| = p - 1 \equiv 2 \pmod 4$, we get $\mathrm{wt}(c_S) \equiv 2|S| \pmod 4$.

(2) By (1), $r_N r_N^{-} + r_Q r_Q^{-} = r_N^2 + r_Q^2 = (r_N + r_Q)^2 = (1+J)^2 = 1 + J$. So $\varphi(u) \perp C_{NQ}$ iff $\mathrm{ct}(u(1+J)v^{-}) = 0$ for all v, iff $u(1+J) = 0$, iff $u \in \{0, J\} = \ker\varphi$. Hence $C_{NQ} \cap C_{NQ}^{\perp} = \{0\}$. □

Remark 3.5. Part (1) is also noted by Boston and Hao [3].

3.2 Weights and point counts

Here we prove the dictionary between codewords and curves: the weight of c_S is determined by the point count of X_S or of X_{S^c}, whichever has even size. Two worked examples follow.

Proposition 3.6. *Let* $S \subseteq \mathbb{F}_p$ *and write* $S^c = \mathbb{F}_p \setminus S$.

(a) If $|S|$ *is even and* $S \neq \emptyset$*:* $\mathrm{wt}(c_S) = p - \Sigma_S = 2p + 2 - \#X_S$.

(b) If $|S|$ *is odd,* $S \neq \mathbb{F}_p$*, and* $p \equiv 1 \pmod 4$*:* $\mathrm{wt}(c_S) = p - \Sigma_{S^c} = 2p + 2 - \#X_{S^c}$.

(c) If $|S|$ *is odd,* $S \neq \mathbb{F}_p$*, and* $p \equiv 3 \pmod 4$*:* $\mathrm{wt}(c_S) = p + \Sigma_{S^c} = \#X_{S^c} - 2$.

In every case the set appearing on the right has even size.

Proof. By (2),

$$p-\mathrm{wt}(c_S)=\sum_{a\in\mathbb{F}_p} T_a(S), \qquad T_a(S)=1-\pi\big(|Q\cap(a-S)|\big)-\pi\big(|N\cap(a-S)|\big).$$

$|S|$ even, $a\in S$. Then $0\in a-S$, so $|Q\cap(a-S)|+|N\cap(a-S)|=|S|-1$ is odd. The two parities differ, so $T_a(S)=0=\chi(f_S(a))$.

$|S|$ even, $a\notin S$. The two counts sum to $|S|$, so they have equal parity. Then $T_a(S)=1$ if both are even and $T_a(S)=-1$ if both are odd. In either case $T_a(S)=(-1)^{|N\cap(a-S)|}=\prod_{s\in S}\chi(a-s)=\chi(f_S(a))$.

So for $|S|$ even, $p-\mathrm{wt}(c_S)=\Sigma_S$, which is (a) by Lemma 2.2.

$|S|$ odd. Since $x\mapsto a-x$ is a bijection, $a-S$ and $a-S^c$ partition $\mathbb{F}_p$. So $|Q\cap(a-S)|+|Q\cap(a-S^c)|=|Q|$, and similarly for N. If $p\equiv 1 \pmod 4$ then $|Q|=|N|$ is even, the parities agree, and $T_a(S)=T_a(S^c)$. If $p\equiv 3 \pmod 4$ they are odd, each parity flips, and $T_a(S)=-T_a(S^c)$. As $|S^c|$ is even, the even case applies to S^c, giving (b) and (c). □

The excluded cases are $c_\emptyset=0$ and $c_{\mathbb{F}_p}$, which is 0 for $p\equiv 1 \pmod 4$ and the all-ones word for $p\equiv 3 \pmod 4$.

Example 3.7. For $p=7$ and $S=\{0,1\}$, Example 2.1 gives $\mathrm{wt}(c_S)=8$ and Example 2.3 gives $\#X_S=8$, so $2p+2-\#X_S=8$, as in (a). For $S=\{0,1,3\}$, $S^c=\{2,4,5,6\}$ and a direct count gives $\#X_{S^c}=12$. Part (c) predicts $\mathrm{wt}(c_S)=10$, which is what one finds.

Example 3.8. Let $p=11$, so $Q=\{1,3,4,5,9\}$. For $S=\{1,2,3,4\}$,

$$c_S=(x^{10}+x^9+x^7+x^6+x^5+x^4+x^2+1,\ \ x^{10}+x^9+x^7+x^6+x^5+x^3+x+1),$$

the vector $(1,0,1,0,1,1,1,1,0,1,1,\ \ 1,1,0,1,0,1,1,1,0,1,1)$ of weight 16. Here $\Sigma_S=-5$ and $\#X_S=8$, and indeed $16=p-\Sigma_S=2p+2-\#X_S$. For $S=\{1,2,3\}$,

$$c_S=(x^9+x^7+x^5+x^4+x^3+x^2+x,\ \ x^{10}+x^8+x^6+x^3+x^2+x+1)$$

has weight 14, and $\Sigma_{S^c}=3$, matching $14=p+\Sigma_{S^c}$.

3.3 The even subcode

Restricting to sets S of even size gives a subcode whose every nonzero weight is $2p+2-\#X_S$. A lower bound on its minimum distance is therefore exactly an upper bound on point counts. This is what drives the bounds in Section 4.

Definition 3.9. Let $C_{\mathrm{ev}}=\{c_S:\ |S| \text{ even}\}=\varphi(R_{\mathrm{ev}})$, where $R_{\mathrm{ev}}=\{u\in R:\mathrm{wt}(u) \text{ even}\}=(1+x)R$.

Proposition 3.10. *For every odd prime p, C_{ev} is a $[2p,\,p-1]$ code, and its nonzero weights are exactly the numbers $2p+2-\#X_T$ for non-empty T of even size. Hence*

$$d(C_{\mathrm{ev}})=2p+2-M_p.$$

Moreover:

1. *if* $p \equiv 1 \pmod 4$ *then* $C_{\mathrm{ev}} = C_{NQ}$*;*

2. *if* $p \equiv 3 \pmod 4$ *then* C_{ev} *is the doubly-even subcode of* C_{NQ}*,* $\#X_T \equiv 0 \pmod 4$ *for every non-empty* T *of even size, and* $d(C_{NQ}) = \min(2p+2-M_p,\ m_p-2)$.

Proof. The kernel of φ is $\{0\}$ or $\{0, J\}$, and J has odd weight, so φ is injective on R_{ev}, which has dimension $p-1$. The weights come from Proposition 3.6(a). If $p \equiv 1 \pmod 4$ then $c_S = c_{S^c}$ (as $\varphi(J) = 0$), and $S \mapsto S^c$ swaps the two parities, so $C_{\mathrm{ev}} = C_{NQ}$. If $p \equiv 3 \pmod 4$, Theorem 3.4(1) identifies C_{ev} with the doubly-even subcode, so $2p+2-\#X_T \equiv 0 \pmod 4$, which gives $\#X_T \equiv 2p+2 \equiv 0 \pmod 4$. The formula for $d(C_{NQ})$ follows from Proposition 3.6(a),(c). □

Example 3.11. For $p = 11$ the even-S codewords of C_{NQ} have weights $8, 12, 16$, with multiplicities $330, 616, 77$. The odd-S codewords have weights $6, 10, 14, 22$, with multiplicities $77, 616, 330, 1$. So $M_{11} = 24-8 = 16$ and $m_{11} = 24-16 = 8$, and $d(C_{NQ}) = \min(8, 6) = 6$ is attained only by odd S. A weight-6 word is c_S with $S = \{0,5,6,7,8,9,10\}$: its complement $T = \{1,2,3,4\}$ has $\#X_T = 8$ (Example 2.5).

3.4 Weight distributions

This subsection records numerical data that illustrate the results above and supply the examples of Section 6. Table 1 lists the weight distributions of C_{NQ} for small p, computed by exhaustive enumeration.

p	nonzero weights w with multiplicities A_w
5	$4^5\ 6^{10}$
7	$4^{14}\ 6^{49}\ 8^{49}\ 10^{14}\ 14^1$
11	$6^{77}\ 8^{330}\ 10^{616}\ 12^{616}\ 14^{330}\ 16^{77}\ 22^1$
13	$8^{273}\ 10^{598}\ 12^{1105}\ 14^{1300}\ 16^{598}\ 18^{182}\ 20^{39}$
17	$6^{136}\ 8^{170}\ 10^{136}\ 12^{4692}\ 14^{11560}\ 16^{16473}\ 18^{16184}\ 20^{10404}\ 22^{4624}\ 24^{1156}$
19	$8^{171}\ 10^{1862}\ 12^{10374}\ 14^{36765}\ 16^{84759}\ 18^{128212}\ 20^{128212}\ 22^{84759}\ 24^{36765}\ 26^{10374}\ 28^{1862}$ $30^{171}\ 38^1$
23	$8^{1012}\ 12^{2576}\ 14^{64009}\ 16^{256542}\ 18^{651728}\ 20^{1303456}\ 22^{1914980}\ 24^{1914980}\ 26^{1303456}\ 28^{651728}$ $30^{256542}\ 32^{64009}\ 34^{2576}\ 38^{1012}\ 46^1$

Table 1: Weight distributions of C_{NQ}; w^A means $A_w = A$.

Remark 3.12. If $p \equiv \pm 1 \pmod 8$, then r_Q and r_N are idempotents, and each of the two components of c_S lies in a binary quadratic residue code of length p. So $C_{NQ} \subseteq \mathrm{QR} \oplus \mathrm{QR}'$, and the square-root bound for QR codes gives $d(C_{NQ}) \geq \sqrt{p}$. Via Proposition 3.6 this yields $\#X_T \geq \sqrt{p}+2$ for $p \equiv 7 \pmod 8$ and non-empty even T. That point-count bound, however, already follows from Lemma 2.2(3) together with (3). If $|T| \geq \sqrt{p}$, then $\#X_T \geq |T|+2$. If $|T| < \sqrt{p}$, then $\#X_T \geq p+1-(|T|-2)\sqrt{p} > 2\sqrt{p}+1$.

4 Point-count bounds from coding bounds

This section applies the dictionary of Section 3 in both directions. A hypothetical upper bound on point counts would produce codes beating the GV bound (Theorem 4.1). Conversely, known upper bounds on codes force the existence of curves with many points: the MRRW bound gives Theorem 4.3, and Rains' bound for self-dual codes gives Corollary 4.5. The section ends by explaining why a longer variant of the code yields nothing new.

Theorem 4.1. *Let $c < 2\big(1 - H^{-1}(1/2)\big) = 1.7799\ldots$ and suppose $B_{\mathrm{ev}}(c,p)$ (that is, $M_p \le cp$) holds for infinitely many primes p. Then Goppa's conjecture 1.1 is false.*

Proof. For such p, Proposition 3.10 makes C_{ev} a $[2p, p-1, \ge (2-c)p+2]$ code. Along these primes the rate tends to $1/2$ and the relative distance is at least $\delta_0 = 1 - c/2 > H^{-1}(1/2) = 0.1100\ldots$. So $\alpha(\delta_0) \ge 1/2 > 1 - H(\delta_0)$. □

Remark 4.2. Working with C_{ev} rather than with all of C_{NQ} is what makes the theorem hold for every prime. When $p \equiv 3 \pmod 4$, the codewords c_S with $|S|$ odd have weights $\#X_{S^c} - 2$, which an upper bound on point counts does not control. Note also that only sets of even size are involved.

Theorem 4.3. *Let $\delta^* = 0.18707\ldots$ be the root in $(0,1/2)$ of $H\big(\frac{1}{2} - \sqrt{\delta(1-\delta)}\big) = \frac{1}{2}$, and let $c^* = 2(1-\delta^*) = 1.62584\ldots$. For every $c < c^*$ there is $p_0(c)$ such that $M_p > cp$ for all primes $p > p_0(c)$. In particular, for all large p there is a set S of even size with $\#X_S > 1.625p$.*

Proof. Otherwise $M_p \le cp$ for infinitely many p. By Proposition 3.10, the codes C_{ev} for those p have rate $\to 1/2$ and relative distance at least $\delta_0 = 1 - c/2 > \delta^*$. The first McEliece–Rodemich–Rumsey–Welch bound [26] (see [16, Thm. 2.10.6]) gives $\limsup R \le H\big(\frac{1}{2} - \sqrt{\delta_0(1-\delta_0)}\big) < \frac{1}{2}$, a contradiction. (The right-hand side is decreasing in δ_0 on $(0,1/2)$.) □

Remark 4.4. The proof gives no way to compute $p_0(c)$ or S. Any improvement of the linear-programming bound at rate $1/2$ improves c^*.

Corollary 4.5. *If $p \equiv 3 \pmod 4$ and $p \ge 11$, then $M_p \ge \frac{5}{3}p - 6$.*

Proof. By Theorem 3.4, C_{NQ} is a Type I self-dual code of length $n = 2p$. Rains' bound [27] gives $d \le 4\lfloor n/24 \rfloor + 4$, or $+6$ if $n \equiv 22 \pmod{24}$; in all cases $d \le p/3 + 6$. By Proposition 3.10, $d = \min(2p + 2 - M_p,\ m_p - 2)$. If $d = 2p + 2 - M_p$ then $M_p \ge \frac{5}{3}p - 4$. Otherwise $m_p \le p/3 + 8$, and Corollary 2.6 gives $M_p \ge 2p + 2 - m_p \ge \frac{5}{3}p - 6$. □

Example 4.6. For $p = 11, 19, 23$ the true values are $M_p = 16, 32, 40$, against $\frac{5}{3}p - 6 = 12.3$, 25.7 and 32.3. For $p = 11$ the minimum weight of C_{NQ} comes from an odd set (Example 3.11). So the second case of the proof, which needs the twist identity, really occurs.

Remark 4.7. One might hope to do better with the *long quadratic residue* (LQR) code

$$L = \{(r_N r_S,\ r_Q r_S,\ r_N r_S^*,\ r_Q r_S^*) :\ S \subseteq \mathbb{F}_p\} \subseteq \mathbb{F}_2^{4p}, \qquad r_S^* = r_S + J.$$

It does not help. Since $r_N r_S^* = r_N r_S + |N| J$, when $p \equiv 1 \pmod 4$ the code L is the doubled code $\{(u,u) : u \in C_{NQ}\}$. When $p \equiv 3 \pmod 4$, L is the coset $(0,0,J,J) + \{(u,u)\}$ of that doubled code, and all of its words have weight $2p$. In both cases the linear span of L has minimum distance $2\,d(C_{NQ})$ and rate about $1/4$. That is the same relative distance as C_{NQ} at half the rate, so L gives nothing that C_{ev} does not.

5 Explicit constructions

The bounds of Section 4 prove that curves with many points exist, but say nothing about how to find them. This section gives explicit sets S built from subgroups of $\mathbb{F}_p^\times$. The results are due to F. Voloch [33], except for the case $p \equiv 5 \pmod 8$ in Theorem 5.1, which is new here. They use no coding theory. The first construction works for every prime and gives about $1.5p$ points. The second gives nearly $2p$ points, but only for primes in a set of positive density.

Theorem 5.1. *Let $p \geq 11$ be prime.*

1. *If $p \equiv 1 \pmod 8$, then $\#X_Q = \frac{3}{2}p + \frac{5}{2}$.*
2. *If $p \equiv 3 \pmod 8$, then $\#X_Q = \frac{3}{2}p - \frac{1}{2}$.*
3. *If $p \equiv 5 \pmod 8$, then $\#X_{Q\cup\{0\}} = \frac{3}{2}p + \frac{1}{2}$.*
4. *If $p \equiv 7 \pmod 8$, then $\#X_N = \frac{3}{2}p + \frac{3}{2}$.*

In particular, every prime $p \geq 11$ has an explicit S with $\#X_S \geq \frac{3}{2}p - \frac{1}{2}$.

Proof. Since $b^{(p-1)/2} = \chi(b)$ for $b \neq 0$, we have $f_Q(x) = x^{(p-1)/2} - 1$ and $f_N(x) = x^{(p-1)/2} + 1$. Write $h = (p-1)/2 = |Q| = |N|$.

(1),(2): f_Q vanishes on Q, $f_Q(n) = -2$ for $n \in N$, and $f_Q(0) = -1$. So $\Sigma_Q = h\chi(-2) + \chi(-1)$. Now $\chi(-2) = 1$ iff $p \equiv 1, 3 \pmod 8$. For $p \equiv 1 \pmod 8$, $\Sigma_Q = h+1$ and $|Q|$ is even, so $\#X_Q = p + 2 + h + 1$. For $p \equiv 3 \pmod 8$, $\Sigma_Q = h - 1$ and $|Q|$ is odd, so $\#X_Q = p + 1 + h - 1$.

(3): With $S = Q \cup \{0\}$, $f_S(x) = x(x^h - 1)$. Then $f_S(n) = -2n$ for $n \in N$, and $\chi(-2n) = \chi(-2)\chi(n) = (-1)(-1) = 1$ when $p \equiv 5 \pmod 8$. Also f_S vanishes on S. So $\Sigma_S = h$, $|S| = h+1$ is odd, and $\#X_S = p + 1 + h$.

(4): $f_N(q) = 2$ for $q \in Q$, $\chi(2) = 1$, $f_N(0) = 1$, and $|N|$ is odd. So $\Sigma_N = h+1$ and $\#X_N = p + 1 + h + 1$. □

Example 5.2. For $p = 13 \equiv 5 \pmod 8$, $Q = \{1, 3, 4, 9, 10, 12\}$ and $S = \{0, 1, 3, 4, 9, 10, 12\}$ gives $\#X_S = 20 = \frac{3}{2} \cdot 13 + \frac{1}{2}$. For $p = 11$, $\#X_Q = 16 = M_{11}$, so here the explicit set is optimal. These values were checked by direct counting for all $p < 200$.

Theorem 5.3 (Voloch)**.** *Fix an integer $\ell \geq 2$. There is a set of primes $p \equiv 1 \pmod \ell$ of positive Dirichlet density such that, with $P_\ell \subset \mathbb{F}_p^\times$ the subgroup of ℓ-th powers,*

$$\#X_{P_\ell} = \Big(2 - \frac{1}{\ell}\Big)p + O(1),$$

where the $O(1)$ term lies in $[-2, 3]$.

Proof. Let ζ be a primitive ℓ-th root of unity in $\mathbb{C}$ and $K = \mathbb{Q}\big(\zeta, \sqrt{\zeta^i - 1} : 1 \leq i \leq \ell - 1\big)$. By Chebotarev's density theorem the primes that split completely in K have density $1/[K : \mathbb{Q}] > 0$; such p satisfy $p \equiv 1 \pmod \ell$. Choose the reduction of ζ to be a primitive ℓ-th root $r \in \mathbb{F}_p$. Then $\chi(r^i - 1) = 1$ for $1 \leq i \leq \ell - 1$. Now $f_{P_\ell}(x) = x^{(p-1)/\ell} - 1$, which takes the value $r^i - 1$ on the coset where $x^{(p-1)/\ell} = r^i$. So every $x \notin P_\ell \cup \{0\}$ contributes $+1$ to Σ_{P_ℓ}. Hence $\Sigma_{P_\ell} = (p-1)(1 - 1/\ell) + \chi(-1)$, and Lemma 2.2 gives the claim. Effective versions of Chebotarev's theorem [22, 29] make such primes, and hence S, computable. □

Corollary 5.4. *Question 1.5 has a negative answer: for every $c < 2$, there are infinitely many primes p (a set of positive density) for which $B(c,p)$ fails.*

This does *not* disprove Conjecture 1.3, which asks for infinitely many primes where $B(c,p)$ *holds*.

6 Duursma zeta functions

This section turns from point counts to a different invariant of the codes, Duursma's zeta function. We recall its definition and the analog of the Riemann hypothesis. We then prove that the extension C' of C_{NQ} is isodual when $p \equiv 1 \pmod 4$, so that the Riemann hypothesis makes sense for it. Finally we tabulate which of our codes satisfy it.

Duursma [9, 10] attached to an $[n,k,d]$ code C over $\mathbb{F}_q$, with dual distance $d^\perp \geq 2$, a zeta function

$$Z_C(T) = \frac{P_C(T)}{(1-T)(1-qT)},$$

where the *zeta polynomial* P_C has degree $n+2-d-d^\perp$, satisfies $P_C(1) = 1$, and depends only on the weight enumerator. Concretely, P_C is the unique polynomial with

$$W_C = \sum_{i=0}^{n-d-d^\perp+2} [T^i] P_C(T) \cdot M_{n,d+i},$$

where $M_{n,w}$ is the weight enumerator of a (formal) MDS code of length n and minimum distance w. With genus $\gamma = n+1-k-d$, the MacWilliams identity becomes the functional equation $P_{C^\perp}(T) = P_C(1/(qT))\, q^\gamma T^{\gamma+\gamma^\perp}$. For formally self-dual C, the zeros of P_C are symmetric about $|T| = q^{-1/2}$, and C is said to satisfy the *Riemann hypothesis* (RH) if all of them lie on $|T| = q^{-1/2}$. Duursma conjectured that every extremal self-dual code satisfies RH [11]. For even binary codes one checks that $d = 2 - \sum_i \rho_i^{-1}$ over the zeros ρ_i of P_C, equivalently $[T^1]P_C/[T^0]P_C = d-2$.

Example 6.1. For $p = 7$, C_{NQ} is a self-dual $[14,7,4]$ code with

$$P(T) = \tfrac{2}{143} + \tfrac{4}{143}T + \tfrac{19}{429}T^2 + \tfrac{28}{429}T^3 + \tfrac{40}{429}T^4 + \tfrac{56}{429}T^5 + \tfrac{76}{429}T^6 + \tfrac{32}{143}T^7 + \tfrac{32}{143}T^8.$$

The coefficients sum to 1, the functional equation $p_8 = 2^4 p_0$ holds, and all eight zeros have modulus $1/\sqrt{2}$.

6.1 The extended code for $p \equiv 1 \pmod 4$

When $p \equiv 1 \pmod 4$ the code C_{NQ} is not self-dual, so the Riemann hypothesis does not directly apply to it. Adjoining the all-ones word repairs this. This subsection proves that the resulting code is equivalent to its dual, works out the case $p = 13$, and collects the results for all our examples in Table 2.

Theorem 6.2. *Let $p \equiv 1 \pmod 4$ and $C' = C_{NQ} + \langle \mathbf{1} \rangle$. Then $\mathbf{1} \notin C_{NQ}$, $\dim C' = p$, and $C'^\perp = \sigma(C')$, where σ swaps the two halves of each codeword. So C' is isodual, hence formally self-dual. Its weight distribution is $A_w + A_{2p-w}$, where (A_w) is that of C_{NQ}.*

Proof. If $(r_N u, r_Q u) = (J, J)$ then $(1+J)u = 0$, so $u \in \{0, J\}$ and $\varphi(u) = 0$, a contradiction. So $\mathbf{1} \notin C_{NQ}$ and $\dim C' = p$. Next, $\sigma(\varphi(v)) = (r_Q v, r_N v)$ and $\langle \varphi(u), \sigma\varphi(v) \rangle = \mathrm{ct}\left(uv^{-}(r_N r_Q^{-} + r_Q r_N^{-})\right) = \mathrm{ct}(uv^{-} \cdot 2 r_N r_Q) = 0$ by (1). The word $\mathbf{1}$ is orthogonal to every even-weight word, and $\langle \mathbf{1}, \mathbf{1} \rangle = 2p = 0$. So $\sigma(C') \subseteq C'^{\perp}$, with equality since both have dimension p. □

Example 6.3. For $p = 13$, C' is a $[26, 13, 6]$ formally self-dual code with weight distribution 1, 39, 455, 1196, 2405, 2405, 1196, 455, 39, 1 in weights $0, 6, 8, \ldots, 20, 26$. Its zeta polynomial has degree 16:

$$P(T) = \tfrac{3}{17710} + \tfrac{6}{8855}T + \tfrac{611}{336490}T^2 + \tfrac{9}{2185}T^3 + \tfrac{3441}{408595}T^4 + \tfrac{6448}{408595}T^5 + \tfrac{44499}{1634380}T^6 + \tfrac{22539}{520030}T^7 + \tfrac{66303}{1040060}T^8 + \tfrac{22539}{260015}T^9 + \tfrac{44499}{408595}T^{10} + \tfrac{51584}{408595}T^{11} + \tfrac{55056}{408595}T^{12} + \tfrac{288}{2185}T^{13} + \tfrac{19552}{168245}T^{14} + \tfrac{768}{8855}T^{15} + \tfrac{384}{8855}T^{16}.$$

Exactly 12 of its 16 zeros have modulus $1/\sqrt{2}$, so C' does not satisfy RH.

The minimum distance 6 is the largest possible for an even binary $[26, 13]$ code if, as we understand the tables [13], no binary $[26, 13, 8]$ code exists. We have not independently verified this, and the claim of optimality should be read with that caveat. Failures of RH for non-extremal self-dual and formally self-dual codes are known; see [11, 6, 7]. The examples here are of interest because the codes are structured and have large minimum distance.

code	p	parameters	zeros on $\lvert T\rvert = 1/\sqrt{2}$	RH
C_{NQ}	7	$[14, 7, 4]$ self-dual	8 of 8	yes
C_{NQ}	11	$[22, 11, 6]$ self-dual	12 of 12	yes
C_{NQ}	19	$[38, 19, 8]$ self-dual	24 of 24	yes
C_{NQ}	23	$[46, 23, 8]$ self-dual	30 of 32	no
C'	13	$[26, 13, 6]$ isodual	12 of 16	no
C'	17	$[34, 17, 6]$ isodual	22 of 24	no
C'	29	$[58, 29, 12]$ isodual	36 of 36	yes

Table 2: The Riemann hypothesis for QQR codes and their extensions.

The case $p = 23$ shows that C_{NQ} need not satisfy RH when $p \equiv 3 \pmod 4$, even though it does for $p = 7$, 11 and 19. Boston and Hao [3] found the same failure independently.

7 Open questions and their difficulty

This section takes stock of what remains open. We first give a heuristic for the true size of the largest point count M_p, then go through the questions and conjectures of Section 1 one at a time. For each, we say what is known and how hard it appears.

A heuristic. Suppose C_{ev} behaved like a random linear code of rate $1/2$. Then its minimum distance would be close to the GV value $2pH^{-1}(1/2)$, and Proposition 3.10 would give $M_p \approx 2\big(1 - H^{-1}(1/2)\big)p \approx 1.78p$. The data in Table 3 are consistent with this. Corollary 2.6 shows that a large M_p and a small m_p go together.

p	7	11	13	17	19	23
M_p	12	16	20	30	32	40
m_p	8	8	8	12	12	16
M_p/p	1.714	1.455	1.538	1.765	1.684	1.739

Table 3: Extreme point counts over even non-empty sets T.

Question 1.4 (explicit S). For $c = 1.5 - o(1)$ it is answered for every $p \geq 11$ by Theorem 5.1. For c close to 2 it is answered on the positive-density sets of primes in Theorem 5.3. For a fixed $c > 1.5$ and *all* large p it is open, and we expect it to be hard. When $p = 2q+1$ is a safe prime, $\mathbb{F}_p^\times$ has no subgroups other than 1, Q and $\mathbb{F}_p^\times$ to exploit. Matching the non-constructive bound of Theorem 4.3 explicitly looks as hard as making the MRRW-type existence argument effective. A cheap experiment worth doing: take S to be a union of cosets of an index-ℓ subgroup. Choosing the cosets is then a linear-algebra problem over $\mathbb{F}_2$ involving the matrix $\left[\chi(r_i - r_j) = -1\right]$ on the ℓ-th roots of unity.

Question 1.5. Answered negatively by Theorem 5.3.

Conjecture 1.3 (Bazzi–Mitter). Open, and very hard. Any $c < 2$ would make the QQR family asymptotically good. That is close in spirit to the long-open question of whether binary quadratic residue codes are asymptotically good. By Corollary 2.6, it would also force every split even-degree curve X_T, of genus up to about $p/2$, to have at least $(2-c)p+2$ points. No known method controls Legendre sums of polynomials whose degree is comparable to p. The heuristic suggests the conjecture holds for every c slightly above 1.78.

The hypothesis of Theorem 4.1. Open. Heuristically it is false, since 1.7799 is exactly the GV threshold. Proving it would refute Goppa's conjecture, and disproving it for this family would require the kind of lower bound on Legendre sums just described.

Riemann hypothesis. The general question for C_{NQ} is settled negatively (Table 2). Whether RH holds whenever C_{NQ} happens to be extremal is a special case of Duursma's conjecture for extremal Type I codes, which is open.

8 Literature

This section places the paper in context. The first part surveys the work this paper builds on, and the second surveys later developments on the same codes and questions. The split at 2006 is only for convenience.

8.1 Before 2006

The background falls into three strands: quadratic residue and double-circulant codes, bounds on codes, and the theory of self-dual codes and their zeta functions.

Quadratic residue codes go back to Prange (1957–1959). Their square-root bound and basic theory are in MacWilliams and Sloane [23] and Huffman and Pless [16]. Double-circulant codes built from quadratic residues were studied by Karlin [20]. Calderbank [4] proved a square-root bound for quasi-cyclic codes. Chen, Peterson and Weldon [5] and Kasami [21] showed that quasi-cyclic codes of rate $1/2$ meet the GV bound, which is the origin of the random-code heuristic of Section 7. Helleseth [14] related QR codewords to Legendre sums, and Helleseth and Voloch [15] combined this with Weil's bound [34] to bound the minimum distance of double-circulant QR codes by about $2\sqrt{p}$. Voloch [32] studied the asymptotics of QR code minimum distances. Bazzi and Mitter [2] (preprint 2001) introduced the QQR codes and Conjecture 1.3.

On bounds: Gilbert (1952) and Varshamov (1957) gave the GV bound. McEliece, Rodemich, Rumsey and Welch [26] gave the best known asymptotic upper bounds. Goppa stated his conjecture in [12]. Jiang and Vardy [17] improved the GV bound on code size by a linear factor, without changing the asymptotic rate.

For self-dual codes: Gleason's theorem and the Mallows–Sloane bound [24] constrain weight enumerators, and Rains [27] sharpened the Type I bound via shadows. Massey [25] introduced LCD codes. Duursma defined zeta functions for codes [9, 10], formulated the Riemann hypothesis analog and studied extremal enumerators [11]. Chinen [6] extended the theory to formal weight enumerators. Character sums and curves over finite fields are treated in Schmidt [28] and Tsfasman and Vlăduţ [31]. Effective Chebotarev theorems are due to Lagarias and Odlyzko [22] and Serre [29].

8.2 2006 and after

Later work has refined the weight distributions of QQR codes, tested the Riemann hypothesis for codes, and studied double-circulant codes in general. None of it settles the central question of whether this specific family is asymptotically good.

A revised version of this paper appeared in 2008 [18]. Joyner and Kim [19] treated the subject at book length, with chapters on the Riemann hypothesis for codes and on hyperelliptic curves and QR codes. Boston and Hao [3] showed that $\mathrm{PSL}_2(p)$ acts on extended QQR codes. They proved divisibility results for QQR weight polynomials, used them to compute weight distributions efficiently, found the $p = 23$ failure of RH, and showed that a symmetrized point-count distribution for a family of the curves X_T tends to the normal distribution. Chinen [7] gave extremal invariant polynomials that violate RH. Alahmadi, Özdemir and Solé [1] showed that self-dual double-circulant codes are dihedral and that the family is asymptotically good, in part assuming the generalized Riemann hypothesis via Artin's primitive root conjecture. Shi, Xu and Solé [30] introduced double Toeplitz codes, which are isodual like our C', and constructed some arbitrarily close to the GV bound at rate $1/2$.

About this revision

This is a revision of the version submitted to *Discrete Mathematics and Theoretical Computer Science* on 20 October 2006. A different revision was published as [18] (see also arXiv:math/0609562). The present text corrects several errors in the 2006 version, replaces some

arguments, proves two statements that were stated as conjectures, and adds examples, a literature survey and a glossary. A separate revision log lists every change.

Acknowledgements

The first author thanks Amin Shokrollahi for helpful advice, Felipe Voloch for allowing his constructions to be included, and his former student Greg Coy, whose honors project [8] contains parts of Proposition 3.6.

This revision was prepared with Claude (Anthropic). Claude refereed the earlier text and proposed and wrote the following, which the first author has reviewed: the proofs of Theorems 3.4 and 6.2; the twist identity (Lemma 2.4) and Corollary 2.6; the use of C_{ev} in Theorems 4.1 and 4.3; the repaired Corollary 4.5; case (3) of Theorem 5.1; and the computations in Tables 1–3. The Python code used to verify every numerical claim is in the repository accompanying this paper. Every change from the earlier text, with the reason for it, is recorded in the separate revision log (`revision-log.tex`) in the same repository.

A Glossary

This glossary collects, in alphabetical order, the terms and notation used in the paper. It is meant for readers who come from only one of the two sides, coding theory or arithmetic geometry. Where a term is defined precisely in the text, the entry points there.

Asymptotically good A family of codes of length $n \to \infty$ whose rate and relative distance both stay bounded away from 0.

$B(c,p)$**,** $B_{\mathrm{ev}}(c,p)$ The statement that $\#X_S \le cp$ for all non-empty $S \subseteq \mathbb{F}_p$ (respectively, all such S of even size); Definition 1.2.

Binary linear code; $[n,k,d]$ A k-dimensional subspace of $\mathbb{F}_2^n$ with minimum nonzero Hamming weight d.

c_S The QQR codeword $(r_N r_S, r_Q r_S)$.

C_{NQ} The quasi-quadratic residue code; Definition 3.1.

C_{ev} The subcode $\{c_S : |S| \text{ even}\}$ of C_{NQ}, of dimension $p-1$; its weights are $2p+2-\#X_T$.

C' For $p \equiv 1 \pmod 4$, the code $C_{NQ} + \langle \mathbf{1} \rangle$; it is isodual (Theorem 6.2).

Character sum Σ_S $\sum_{x \in \mathbb{F}_p} \chi(f_S(x))$; it determines $\#X_S$ by Lemma 2.2.

Chebotarev density theorem For a finite Galois extension $K/\mathbb{Q}$, the primes whose Frobenius lies in a given conjugacy class have natural density proportional to the size of that class. In particular, the primes that split completely have density $1/[K:\mathbb{Q}]$.

Double-circulant code A code with generator matrix $[A \mid B]$, where A, B are $p \times p$ circulant matrices; a quasi-cyclic code of index 2.

Doubly-even, singly-even A binary code is doubly-even if all its weights are divisible by 4, and singly-even if it is even but not doubly-even.

Dual code $C^\perp$ $\{v : \langle v,c\rangle = 0 \ \forall c \in C\}$ with the standard inner product.

Duursma zeta function, zeta polynomial $Z_C(T) = P_C(T)/((1-T)(1-qT))$, where P_C is determined by the weight enumerator; Section 6.

Entropy function H $H(\delta) = -\delta\log_2\delta - (1-\delta)\log_2(1-\delta)$.

Extremal code A self-dual code whose minimum distance meets the relevant upper bound (Mallows–Sloane for Type II, Rains for Type I).

f_S The monic polynomial $\prod_{a\in S}(x-a) \in \mathbb{F}_p[x]$.

Formally self-dual A code with the same weight enumerator as its dual.

Genus of a code Duursma's $\gamma = n+1-k-d$, which measures the distance from the Singleton bound.

Genus of X_S $\lfloor (|S|-1)/2 \rfloor$.

Gilbert–Varshamov (GV) bound The guarantee of codes with $k \geq n - \log_2 V(n,d-1)$; asymptotically, rate $1-H(\delta)$.

Goppa's conjecture The conjecture that the asymptotic GV bound is tight for binary codes; Conjecture 1.1.

Hamming weight, distance $\mathrm{wt}(v)$ is the number of nonzero coordinates of v, and $d(u,v) = \mathrm{wt}(u-v)$.

Hyperelliptic curve X_S The smooth projective curve with affine equation $y^2 = f_S(x)$.

Idempotent An element e of R with $e^2 = e$; it generates a cyclic code.

Isodual Equivalent to its own dual by a coordinate permutation. Isodual codes are formally self-dual.

J The all-ones polynomial $1+x+\cdots+x^{p-1} = r_{\mathbb{F}_p}$ in R.

LCD code A code C with $C \cap C^\perp = \{0\}$ (linear complementary dual).

Legendre symbol χ $\chi(b) = b^{(p-1)/2} \in \{0, \pm 1\}$ in $\mathbb{F}_p$.

LQR code The long quadratic residue code of length $4p$; Remark 4.7.

M_p, m_p The maximum and minimum of $\#X_T$ over non-empty T of even size.

MacWilliams identity The relation $W_{C^\perp}(x,y) = |C|^{-1} W_C(x+y, x-y)$ between weight enumerators.

MDS code A code meeting the Singleton bound $d = n-k+1$.

MRRW bound The McEliece–Rodemich–Rumsey–Welch linear-programming upper bounds; the first one is $R \leq H(\frac{1}{2} - \sqrt{\delta(1-\delta)})$.

$\#X_S$ The number of $\mathbb{F}_p$-points of the smooth projective model of X_S, including the points at infinity.

N, Q The non-squares and the nonzero squares in $\mathbb{F}_p^\times$.

Optimal code A code whose minimum distance is the largest possible for its length and dimension, within a stated class of codes.

Points at infinity The points of the smooth model of $y^2 = f(x)$ not in the affine plane. There are two when $\deg f$ is even and f is monic, and one when $\deg f$ is odd.

QR code A binary cyclic code of prime length $p \equiv \pm 1 \pmod 8$ generated by an idempotent built from r_Q or r_N.

Quadratic twist For $y^2 = h(x)$, the curve $y^2 = \nu h(x)$ with ν a non-square. The two point counts sum to $2p+2$ when $\deg h$ is even.

Quasi-cyclic code A code invariant under a simultaneous cyclic shift of blocks of coordinates.

R The ring $\mathbb{F}_2[x]/(x^p-1)$.

r_S The polynomial $\sum_{i \in S} x^i \in R$.

Rains bound For singly-even self-dual codes of length n, $d \leq 4\lfloor n/24 \rfloor + 4$, or $+6$ if $n \equiv 22 \pmod{24}$.

Rate, relative distance For an $[n,k,d]$ code these are k/n and d/n.

Riemann hypothesis (for codes) For a formally self-dual code over $\mathbb{F}_q$, the property that all zeros of P_C lie on $|T| = q^{-1/2}$.

Safe prime A prime p such that $(p-1)/2$ is also prime.

Self-dual, self-orthogonal $C = C^\perp$, respectively $C \subseteq C^\perp$.

Shadow For a singly-even self-dual code C with doubly-even subcode C_0, the set $C_0^\perp \setminus C$. Its weight enumerator constrains that of C.

Type I, Type II Singly-even, respectively doubly-even, binary self-dual codes.

Weil bound $|\#X(\mathbb{F}_q) - (q+1)| \leq 2g\sqrt{q}$ for a smooth projective curve of genus g.

Weight distribution, weight enumerator The numbers A_w of codewords of weight w, and $W_C(x,y) = \sum_w A_w x^{n-w} y^w$.

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